Wu 2020 Acute Myeloid Leukemia — analysis summary

RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

36 AML samples intermix across one continuous manifold; iLISI 9.4 of 23.6 ceiling (θ=10 peak, from 1.1 pre-Harmony), single connected component, one low-cycling compartment.

Gate 2 — module geography → GO (non-branching)

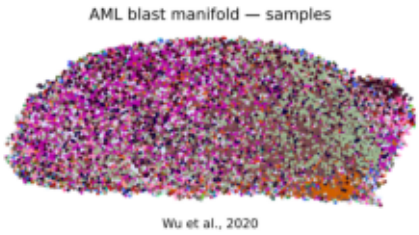
Single continuous Gast/pSom axis with organized cross-repression (smoothed r=−0.68) and a terminal Crest gradient (ρ=+0.43 vs pseudotime). Non-branching case → passes.

Gate 3 — hub cross-repression → NA

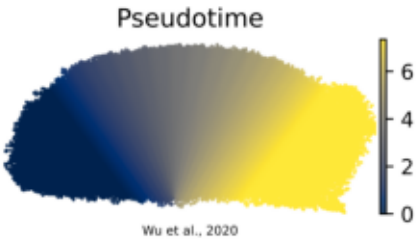
No bifurcation: the PPT tree's tips/fork carry near-identical signature means (geometric, not biological), so there is no co-expression hub or opposing arms to test. Branching-only gate not applicable.

STEP FIGURES

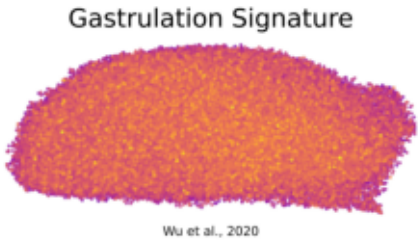
Manifold — samples



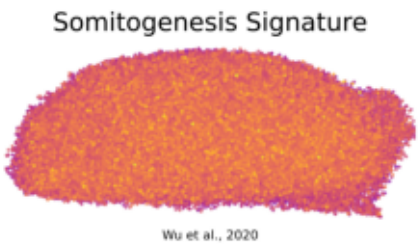
Pseudotime



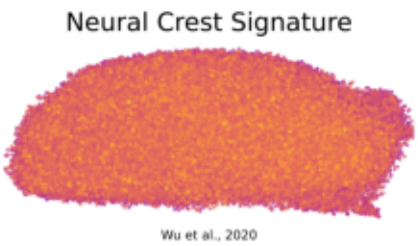
Gastrulation



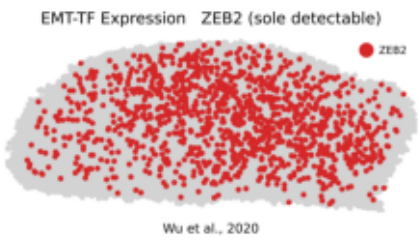
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Acute myeloid leukemia (Wu et al., 2020), Microwell-seq — 96,881 malignant blasts from 36 bone-marrow samples.

Verdict.
The embryonic-development pattern reproduces in its non-branching form — a single continuous gastrulation/somitogenesis axis under organized cross-repression, resolving into a terminal neural-crest gradient. No competing arms (Gate 3 NA). Geometry: Linear-pSom/Crest. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means Gast +0.0019 / pSom +0.0066 / Crest −0.0042. EMT-TF ZEB2 (sole detectable factor; epithelial Snail/Zeb1/Twist silent in blasts).

Parameter choices

Min-cell threshold	202 (viable-range cliff 417→98, geo-mean); 36/38 samples kept	**Integration**	Harmony θ=10 (iLISI sweep peak; drops at θ=20)
Embedding	n_pcs=20, n_neighbors=30, spread=1.3	**Geometry**	principal curve (ElPiGraph); fold-free ratio 2.90; continuity DC1 0.92
Root	curve tip 0 = Gast/pSom-high, Crest-low (origin)	**Module means**	Gast +0.0019 / pSom +0.0066 / Crest −0.0042
EMT-TF	ZEB2 (sole detectably-expressed factor; Snail/Zeb1/Twist silent)		